# 1. Quality Control Analyst

### Job Summary

The Quality Control Analyst is responsible for performing laboratory testing of raw materials, in-process samples, and finished products to ensure compliance with quality standards and regulatory guidelines.

### Key Responsibilities

- Perform chemical and/or microbiological testing as per SOPs
- Analyze raw materials, intermediates, and finished products
- Operate analytical instruments (HPLC, GC, UV, FTIR)
- Record and review test data accurately
- Follow Good Manufacturing Practices (GMP) and Good Laboratory Practices (GLP)
- Report deviations, out-of-specification (OOS) results
- Assist in method validation and stability studies

### Qualifications

- B.Sc / M.Sc in Chemistry, Microbiology, Biotechnology, or related field

### Skills Required

- Analytical and problem-solving skills
- Knowledge of laboratory instruments
- Attention to detail
- Basic understanding of GMP/GLP

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# 2. Junior QC Analyst

### Job Summary

The Junior QC Analyst supports senior analysts in laboratory testing, documentation, and routine quality checks.

### Key Responsibilities

- Assist in sample preparation and testing
- Maintain laboratory records and logbooks
- Calibrate instruments under supervision
- Follow SOPs and safety guidelines
- Support stability testing and documentation

### Qualifications

- B.Sc in Science (Chemistry/Biology preferred)

### Skills Required

- Basic lab knowledge
- Willingness to learn
- Teamwork and discipline

---

# 3. QC Executive

### Job Summary

The QC Executive oversees daily quality control activities, ensuring that testing is completed on time and meets compliance standards.

### Key Responsibilities

- Plan and execute QC testing activities
- Review test results and reports
- Ensure compliance with GMP and regulatory requirements
- Coordinate with production and QA teams
- Handle deviations and investigations

### Qualifications

- B.Sc / M.Sc with 1–3 years experience (or strong fresher)

### Skills Required

- Leadership and coordination
- Knowledge of regulatory guidelines
- Problem-solving ability

---

# 4. QC Chemist

### Job Summary

The QC Chemist performs chemical analysis of substances to ensure product quality and safety.

### Key Responsibilities

- Conduct chemical testing and analysis

- Operate instruments like HPLC, GC, UV
- Prepare reagents and standards
- Maintain lab documentation
- Ensure compliance with testing protocols

### Qualifications

- B.Sc / M.Sc in Chemistry

### Skills Required

- Strong chemistry fundamentals
- Instrument handling
- Data interpretation

---

# 5. QC Microbiologist

### Job Summary

The QC Microbiologist is responsible for microbiological testing to ensure products are free from contamination.

### Key Responsibilities

- Perform microbial testing (sterility, endotoxin, bioburden)
- Conduct environmental monitoring
- Maintain aseptic conditions in the lab
- Prepare culture media and reagents
- Document results and deviations

### Qualifications

- B.Sc / M.Sc in Microbiology / Biotechnology

### Skills Required

- Aseptic techniques
- Knowledge of microbiological methods
- Attention to detail

---

# 6. Laboratory Analyst

### Job Summary

The Laboratory Analyst conducts routine lab tests and supports quality control processes.

### Key Responsibilities

- Perform routine analysis of samples
- Maintain lab equipment and cleanliness
- Document test results accurately
- Follow SOPs and safety procedures

### Qualifications

- B.Sc in relevant scientific field

### Skills Required

- Basic lab skills
- Accuracy and consistency
- Time management

---

# 7. Quality Assistant

### Job Summary

The Quality Assistant supports QC and QA teams in documentation, coordination, and compliance activities.

### Key Responsibilities

- Maintain quality records and documentation
- Assist in audits and inspections
- Track deviations and CAPA records
- Coordinate between departments

### Qualifications

- B.Sc / Any Science Graduate

### Skills Required

- Documentation skills
- Organizational ability
- Basic GMP knowledge

---

# 8. Trainee – Quality Control

## Job Summary

The QC Trainee undergoes structured training in laboratory testing, documentation, and quality systems.

## Key Responsibilities

- Learn laboratory techniques and SOPs
- Assist in testing under supervision
- Maintain training records
- Follow safety and compliance guidelines

## Qualifications

- B.Sc / M.Sc (Freshers)

## Skills Required

- Learning attitude
- Discipline and adaptability
- Basic scientific knowledge

# 9. Analytical Chemist

## Job Summary

The Analytical Chemist is responsible for performing advanced chemical analysis using sophisticated instruments to ensure product quality, safety, and regulatory compliance.

## Key Responsibilities

- Perform qualitative and quantitative chemical analysis
- Develop and execute analytical test methods
- Operate instruments such as HPLC, GC, UV-Vis, FTIR
- Analyze raw materials, intermediates, and finished products
- Interpret analytical data and prepare reports
- Support method validation and transfer activities
- Investigate out-of-specification (OOS) results

## Qualifications

- M.Sc / B.Sc in Chemistry or Analytical Chemistry

### Skills Required

- Strong analytical and problem-solving skills
- Expertise in chromatography and spectroscopy
- Data interpretation and documentation

---

# 10. Senior QC Analyst

### Job Summary

The Senior QC Analyst leads complex testing activities, reviews analytical data, and ensures compliance with regulatory and quality standards.

### Key Responsibilities

- Perform and supervise advanced analytical testing
- Review and approve laboratory data and reports
- Train and mentor junior analysts
- Handle OOS, deviations, and investigations
- Ensure compliance with GMP, GLP, and regulatory guidelines
- Support audits and inspections
- Participate in method validation and stability studies

### Qualifications

- M.Sc / B.Sc with 3–6 years of experience

### Skills Required

- Leadership and mentoring
- Strong knowledge of GMP/GLP
- Advanced analytical skills
- Attention to detail

---

# 11. Instrumentation Analyst (HPLC, GC, UV, FTIR)

### Job Summary

The Instrumentation Analyst specializes in operating, calibrating, and troubleshooting analytical instruments used in quality control laboratories.

### Key Responsibilities

- Operate and maintain instruments such as HPLC, GC, UV, FTIR
- Perform calibration and system suitability testing
- Troubleshoot instrument issues and ensure uptime
- Execute analytical methods using instrumentation
- Maintain instrument logs and documentation
- Coordinate with service engineers for maintenance

### Qualifications

- B.Sc / M.Sc in Chemistry or Instrumentation

### Skills Required

- Hands-on experience with analytical instruments
- Technical troubleshooting
- Knowledge of calibration and validation

---

# 12. Wet Chemistry Analyst

### Job Summary

The Wet Chemistry Analyst performs classical chemical analysis techniques to evaluate the quality of materials and products.

### Key Responsibilities

- Conduct titrations, gravimetric and volumetric analysis
- Perform pH, moisture, and assay testing
- Prepare reagents and standard solutions
- Maintain lab records and documentation
- Follow SOPs and safety protocols
- Support routine quality control testing

### Qualifications

- B.Sc / M.Sc in Chemistry

### Skills Required

- Strong fundamentals in chemistry
- Precision in manual testing methods

- Attention to detail

---

# 13. Stability Testing Analyst

### Job Summary

The Stability Testing Analyst evaluates product stability over time under various environmental conditions to ensure shelf-life and regulatory compliance.

### Key Responsibilities

- Conduct stability studies as per ICH guidelines
- Analyze samples at defined intervals
- Maintain stability chambers and monitoring systems
- Record and trend stability data
- Prepare stability reports and summaries
- Support regulatory submissions

### Qualifications

- B.Sc / M.Sc in Chemistry, Pharmacy, or related field

### Skills Required

- Knowledge of stability protocols and guidelines
- Data analysis and trending
- Documentation and compliance skills

# 14. QC Microbiologist

### Job Summary

The QC Microbiologist is responsible for performing microbiological testing to ensure that products meet safety, sterility, and regulatory standards.

### Key Responsibilities

- Perform microbial limit tests, sterility testing, and bioburden analysis
- Conduct culture preparation, incubation, and colony counting

- Identify microorganisms using biochemical or automated methods
- Maintain aseptic conditions in microbiology labs
- Record and review microbiological data
- Investigate out-of-specification (OOS) results
- Ensure compliance with GMP, GLP, and aseptic practices

### Qualifications

- B.Sc / M.Sc in Microbiology, Biotechnology, or Life Sciences

### Skills Required

- Strong knowledge of microbiological techniques
- Aseptic handling and contamination control
- Attention to detail and documentation skills

---

# 15. Environmental Monitoring Analyst

### Job Summary

The Environmental Monitoring (EM) Analyst is responsible for monitoring and controlling microbiological contamination in cleanrooms and controlled environments.

### Key Responsibilities

- Perform air, surface, and personnel monitoring (settle plates, swabs, contact plates)
- Monitor cleanroom environments (Grade A, B, C, D areas)
- Analyze environmental samples for microbial contamination
- Maintain EM records and trend data
- Report excursions and deviations
- Support cleanroom validation and qualification activities

### Qualifications

- B.Sc / M.Sc in Microbiology or related field

### Skills Required

- Knowledge of cleanroom classifications
- Sampling techniques and contamination control
- Data trending and reporting

---

# 16. Sterility Assurance Analyst

### Job Summary

The Sterility Assurance Analyst ensures that sterile products and manufacturing processes meet sterility requirements and regulatory standards.

### Key Responsibilities

- Perform sterility testing using membrane filtration or direct inoculation methods
- Monitor aseptic manufacturing processes
- Validate sterilization methods (autoclave, dry heat, filtration)
- Conduct media fill (process simulation) studies
- Investigate sterility failures and contamination events
- Ensure compliance with aseptic processing guidelines

### Qualifications

- B.Sc / M.Sc in Microbiology / Biotechnology

### Skills Required

- Expertise in aseptic techniques
- Knowledge of sterilization methods
- Strong investigation and problem-solving skills

---

# 17. Endotoxin Testing Analyst

### Job Summary

The Endotoxin Testing Analyst is responsible for detecting bacterial endotoxins in pharmaceutical and biological products to ensure product safety.

### Key Responsibilities

- Perform endotoxin testing using LAL (Limulus Amebocyte Lysate) method
- Conduct gel clot, turbidimetric, or chromogenic assays
- Prepare standards, reagents, and controls
- Maintain endotoxin testing equipment and records
- Analyze and interpret test results
- Ensure compliance with pharmacopeial standards (USP, EP, IP)

### Qualifications

- B.Sc / M.Sc in Microbiology / Biotechnology / Life Sciences

### Skills Required

- Knowledge of endotoxin testing methods
- Precision and accuracy in lab work
- Understanding of regulatory requirements

# 18. Method Development Analyst

### □ Job Summary

The Method Development Analyst is responsible for developing, optimizing, and standardizing analytical methods for testing raw materials, intermediates, and finished products.

### Key Responsibilities

- Develop analytical methods using HPLC, GC, UV, FTIR
- Optimize parameters (mobile phase, flow rate, wavelength, etc.)
- Conduct method trials and feasibility studies
- Document method development reports
- Collaborate with R&D and QC teams
- Ensure methods meet regulatory requirements

### Qualifications

- M.Sc / B.Sc in Analytical Chemistry or related field

### Skills Required

- Strong analytical and experimental skills
- Instrument expertise
- Problem-solving and method optimization

---

# 19. Method Validation Specialist

### Job Summary

The Method Validation Specialist ensures that analytical methods are reliable, reproducible, and compliant with regulatory standards.

### Key Responsibilities

- Perform method validation as per ICH guidelines (Q2)
- Evaluate parameters: accuracy, precision, specificity, linearity, robustness
- Prepare validation protocols and reports
- Conduct method transfer between labs
- Maintain validation documentation
- Support audits and regulatory inspections

### Qualifications

- M.Sc / B.Sc in Chemistry / Pharmacy

### Skills Required

- Knowledge of ICH Q2 guidelines
- Statistical analysis and data interpretation
- Documentation and compliance expertise

---

# 20. Stability Study Scientist

### Job Summary

The Stability Study Scientist designs and manages stability programs to determine product shelf-life under various environmental conditions.

### Key Responsibilities

- Design stability study protocols (accelerated, long-term, intermediate)
- Store samples in stability chambers (ICH conditions)
- Analyze samples at defined intervals
- Trend and interpret stability data
- Prepare stability reports for regulatory submissions
- Coordinate with QA and regulatory teams

### Qualifications

- M.Sc / B.Pharm / M.Pharm or related field

### Skills Required

- Knowledge of ICH stability guidelines (Q1A)
- Data analysis and reporting
- Attention to detail

---

# 21. Raw Material QC Analyst

### Job Summary

The Raw Material QC Analyst ensures the quality and compliance of incoming raw materials before use in production.

### Key Responsibilities

- Test raw materials against specifications
- Perform identity, purity, and quality checks
- Review supplier certificates of analysis (CoA)
- Approve or reject materials based on results
- Maintain documentation and records
- Support vendor audits and qualification

### Qualifications

- B.Sc / M.Sc in Chemistry or related field

### Skills Required

- Knowledge of raw material specifications
- Analytical testing skills
- Documentation and compliance

---

# 22. In-Process Quality Control (IPQC) Analyst

### Job Summary

The IPQC Analyst monitors and controls product quality during manufacturing to ensure consistency and compliance.

### Key Responsibilities

- Perform in-process checks during production
- Monitor parameters like weight, hardness, pH, viscosity, etc.
- Ensure adherence to batch manufacturing records (BMR)
- Identify deviations and report issues
- Coordinate with production and QA teams
- Maintain IPQC logs and reports

### Qualifications

- B.Sc / B.Pharm / Diploma in relevant field

### Skills Required

- Knowledge of manufacturing processes
- Attention to detail
- Real-time decision-making

---

# 23. Finished Product Testing Analyst

### Job Summary

The Finished Product Testing Analyst ensures that final products meet all quality specifications before release to the market.

### Key Responsibilities

- Perform final product testing (assay, dissolution, content uniformity)
- Review test results against specifications
- Prepare certificates of analysis (CoA)
- Support batch release process
- Investigate out-of-specification results
- Maintain documentation and compliance

### Qualifications

- B.Sc / M.Sc / B.Pharm

### Skills Required

- Analytical testing expertise
- Knowledge of product specifications
- Documentation and reporting

---

# 24. Packaging Quality Analyst

### Job Summary

The Packaging Quality Analyst ensures that packaging materials and processes meet quality standards and regulatory requirements.

### Key Responsibilities

- Inspect packaging materials (labels, cartons, containers)
- Verify packaging compliance with specifications
- Check labeling accuracy and regulatory information
- Perform packaging validation checks
- Monitor packaging operations
- Report defects and deviations

### Qualifications

- B.Sc / Diploma / Engineering (Packaging / Printing / Materials)

### Skills Required

- Knowledge of packaging materials and standards
- Attention to detail
- Quality inspection skills

# 25. In-Process Quality Control (IPQC) Officer

### Job Summary

The IPQC Officer monitors product quality during manufacturing processes to ensure compliance with predefined specifications and regulatory standards.

### Key Responsibilities

- Perform in-process checks at various production stages
- Monitor critical parameters (weight, hardness, thickness, pH, viscosity, etc.)
- Verify adherence to Batch Manufacturing Records (BMR)
- Identify and report deviations or non-conformances
- Approve or reject intermediate products
- Coordinate with production and QA teams
- Ensure GMP compliance on the shop floor

### Qualifications

- B.Sc / B.Pharm / Diploma in Pharmacy or relevant field

### Skills Required

- Knowledge of manufacturing processes
- Attention to detail
- Decision-making in real-time
- Communication and coordination

---

# 26. Line QC Inspector

### Job Summary

The Line QC Inspector is responsible for inspecting products and processes directly on the production line to ensure quality standards are met.

### Key Responsibilities

- Inspect products at different production stages
- Verify labeling, packaging, and physical appearance
- Conduct visual and dimensional checks
- Identify defects and non-conforming products
- Maintain inspection records and reports
- Ensure compliance with SOPs and quality standards

### Qualifications

- Diploma / B.Sc in relevant field

### Skills Required

- Strong observation skills
- Knowledge of inspection techniques
- Attention to detail
- Basic documentation skills

---

# 27. Shop Floor Quality Analyst

### Job Summary

The Shop Floor Quality Analyst ensures that manufacturing activities are performed according to quality standards and regulatory requirements.

### Key Responsibilities

- Monitor production processes in real-time

- Ensure compliance with SOPs and GMP guidelines
- Verify equipment cleanliness and line clearance
- Support process validation activities
- Record and report deviations
- Coordinate with production, QC, and QA teams

### Qualifications

- B.Sc / B.Pharm / Engineering (relevant field)

### Skills Required

- Strong understanding of GMP
- Process monitoring skills
- Problem-solving ability
- Team coordination

---

# 28. Batch Release Analyst

### Job Summary

The Batch Release Analyst reviews production and quality data to ensure that finished batches meet all quality requirements before release to the market.

### Key Responsibilities

- Review Batch Manufacturing Records (BMR) and Batch Packing Records (BPR)
- Verify test results from QC laboratories
- Ensure compliance with specifications and regulatory guidelines
- Approve or reject batches for release
- Coordinate with QA and regulatory teams
- Maintain batch release documentation

### Qualifications

- B.Pharm / M.Pharm / B.Sc / M.Sc

### Skills Required

- Strong documentation and review skills
- Knowledge of GMP and regulatory requirements
- Attention to detail
- Analytical thinking

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# 29. Quality Control Supervisor

### Job Summary

The QC Supervisor oversees quality control operations on the manufacturing floor, ensuring compliance, efficiency, and team performance.

### Key Responsibilities

- Supervise QC inspectors and analysts
- Allocate tasks and manage daily QC activities
- Ensure adherence to GMP, SOPs, and safety standards
- Review inspection and testing reports
- Handle deviations, CAPA, and investigations
- Train and mentor team members
- Support audits and regulatory inspections

### Qualifications

- B.Sc / M.Sc / B.Pharm with 4–8 years of experience

### Skills Required

- Leadership and team management
- Strong GMP knowledge
- Decision-making and problem-solving
- Communication and reporting

# 30. QC Documentation Specialist

### Job Summary

The QC Documentation Specialist manages and controls all quality control documents to ensure accuracy, traceability, and compliance with regulatory standards.

### Key Responsibilities

- Prepare, review, and maintain QC documents (SOPs, STPs, reports)
- Ensure proper documentation practices (GDP – Good Documentation Practices)
- Control document lifecycle (creation, revision, archival, retrieval)
- Maintain logbooks, registers, and documentation systems

- Ensure documents comply with GMP and regulatory requirements
- Support audits by providing required documentation

### Qualifications

- B.Sc / M.Sc / B.Pharm or related field

### Skills Required

- Strong documentation and writing skills
- Knowledge of GMP and GDP
- Attention to detail
- Organizational skills

---

# 31. Data Integrity Analyst

### Job Summary

The Data Integrity Analyst ensures that all quality control data is accurate, complete, and compliant with regulatory expectations (ALCOA+ principles).

### Key Responsibilities

- Monitor and audit QC data for integrity and compliance
- Ensure adherence to ALCOA+ principles (Attributable, Legible, Contemporaneous, Original, Accurate)
- Identify data discrepancies and investigate issues
- Implement data integrity controls and procedures
- Review electronic systems (LIMS, CDS) for compliance
- Support regulatory inspections and audits

### Qualifications

- B.Sc / M.Sc / B.Pharm / IT-related background

### Skills Required

- Knowledge of data integrity principles
- Analytical and auditing skills
- Familiarity with electronic systems (LIMS, ERP)
- Attention to detail

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# 32. Quality Systems Analyst

### Job Summary

The Quality Systems Analyst manages and improves quality management systems (QMS) to ensure regulatory compliance and operational excellence.

### Key Responsibilities

- Maintain and monitor QMS elements (Deviations, CAPA, Change Control)
- Track and analyze quality metrics and KPIs
- Ensure SOP compliance across departments
- Support implementation of quality improvements
- Coordinate training on quality systems
- Prepare reports for management review

### Qualifications

- B.Sc / M.Sc / B.Pharm / Engineering

### Skills Required

- Knowledge of QMS and regulatory frameworks
- Data analysis and reporting
- Problem-solving skills
- Cross-functional coordination

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# 33. Compliance Analyst (QC)

### Job Summary

The Compliance Analyst ensures that QC activities adhere to internal policies and external regulatory standards.

### Key Responsibilities

- Review QC processes for compliance with GMP, GLP
- Conduct internal audits and gap assessments
- Ensure adherence to SOPs and regulatory guidelines
- Track and resolve compliance issues
- Prepare compliance reports and documentation
- Support regulatory submissions and inspections

### Qualifications

- B.Pharm / M.Pharm / B.Sc / M.Sc

### Skills Required

- Strong understanding of regulatory requirements
- Auditing and inspection readiness
- Analytical thinking
- Documentation and reporting

---

# 34. Audit Support Analyst

### Job Summary

The Audit Support Analyst coordinates and supports internal and external audits by ensuring readiness, documentation availability, and compliance.

### Key Responsibilities

- Prepare audit documentation and records
- Coordinate audit schedules and logistics
- Support internal and external audits (USFDA, WHO, ISO)
- Address audit observations and track CAPA
- Maintain audit reports and follow-up actions
- Ensure continuous audit readiness

### Qualifications

- B.Sc / M.Sc / B.Pharm or related field

### Skills Required

- Audit coordination and communication
- Documentation management
- Knowledge of regulatory inspections
- Attention to detail

# 35. Bioanalytical Scientist

### Job Summary

The Bioanalytical Scientist performs quantitative and qualitative analysis of biological samples to support product quality, safety, and regulatory compliance.

### Key Responsibilities

- Conduct bioanalytical testing (PK/PD assays, biomarker analysis)
- Perform assays using ELISA, LC-MS/MS, ligand-binding methods
- Analyze biological samples (plasma, serum, tissues)
- Validate bioanalytical methods as per regulatory guidelines
- Interpret data and prepare scientific reports
- Support clinical and preclinical studies

### Qualifications

- M.Sc / PhD in Biotechnology, Biochemistry, Life Sciences

### Skills Required

- Strong bioanalytical techniques
- Data analysis and interpretation
- Knowledge of regulatory guidelines (GLP, GCP)

---

# 36. Cell Culture QC Analyst

### Job Summary

The Cell Culture QC Analyst is responsible for maintaining, testing, and monitoring cell lines to ensure quality and contamination-free cultures.

### Key Responsibilities

- Maintain mammalian or microbial cell cultures
- Perform cell viability and growth analysis
- Conduct contamination testing (mycoplasma, bacteria, fungi)
- Monitor cell morphology and characteristics
- Prepare media and reagents
- Maintain aseptic conditions in cell culture labs

### Qualifications

- B.Sc / M.Sc in Biotechnology, Microbiology, Life Sciences

### Skills Required

- Aseptic techniques and cell handling
- Knowledge of cell biology

- Lab maintenance and documentation

---

# 37. Molecular Biology QC Analyst

### Job Summary

The Molecular Biology QC Analyst performs genetic and molecular-level testing to ensure product identity, purity, and quality.

### Key Responsibilities

- Perform PCR, qPCR, RT-PCR, and DNA/RNA analysis
- Conduct gel electrophoresis and sequencing
- Analyze genetic material for identity and contamination
- Validate molecular testing methods
- Maintain lab records and documentation
- Support research and development activities

### Qualifications

- B.Sc / M.Sc in Molecular Biology, Biotechnology, Genetics

### Skills Required

- Molecular biology techniques
- Data analysis and interpretation
- Attention to detail

---

# 38. Virology QC Specialist

### Job Summary

The Virology QC Specialist is responsible for detecting and analyzing viral contaminants in biological products to ensure safety and regulatory compliance.

### Key Responsibilities

- Perform viral detection assays (PCR, cell-based assays)
- Conduct viral clearance and inactivation studies
- Monitor viral contamination in biologics production
- Maintain biosafety protocols (BSL labs)
- Document test results and investigations
- Support vaccine and biologics development

### Qualifications

- M.Sc / PhD in Virology, Microbiology, Biotechnology

### Skills Required

- Knowledge of virology and viral assays
- Biosafety handling (BSL-2/BSL-3)
- Analytical and investigative skills

---

# 39. Immunoassay Analyst (ELISA, PCR)

### Job Summary

The Immunoassay Analyst performs immunological and molecular assays to detect and quantify biological molecules for quality control and research purposes.

### Key Responsibilities

- Perform ELISA, Western blot, and PCR-based assays
- Detect proteins, antibodies, and antigens
- Prepare reagents, standards, and controls
- Analyze assay data and generate reports
- Validate assay methods and ensure reproducibility
- Maintain laboratory documentation

### Qualifications

- B.Sc / M.Sc in Biotechnology, Immunology, Life Sciences

### Skills Required

- Immunoassay techniques
- Precision and accuracy in lab work
- Data interpretation

# 40. Digital QC Analyst

### Job Summary

The Digital QC Analyst leverages digital tools and systems to optimize quality control processes, improve efficiency, and ensure data accuracy in laboratory operations.

### Key Responsibilities

- Implement and manage digital QC systems and workflows
- Monitor electronic data from laboratory instruments
- Ensure compliance with data integrity (ALCOA+) principles
- Analyze digital QC data for trends and anomalies
- Support digital transformation initiatives in QC labs
- Coordinate with IT and QA teams for system integration

### Qualifications

- B.Sc / M.Sc in Science, IT, or related field

### Skills Required

- Knowledge of LIMS, CDS, and digital tools
- Data analysis and visualization
- Understanding of GMP and data integrity

---

# 4. Laboratory Information Management System (LIMS) Specialist

### Job Summary

The LIMS Specialist manages and optimizes Laboratory Information Management Systems to ensure efficient data handling, workflow automation, and compliance.

### Key Responsibilities

- Configure and maintain LIMS software
- Manage sample tracking, test workflows, and reporting
- Ensure system validation and compliance (CSV – Computer System Validation)
- Troubleshoot system issues and provide user support
- Integrate LIMS with lab instruments and other systems
- Train users on LIMS functionalities

### Qualifications

- B.Sc / M.Sc / B.Tech in IT, Biotechnology, or related field

### Skills Required

- LIMS platforms (LabWare, STARLIMS, Veeva)
- SQL/database knowledge
- System validation (CSV)
- Problem-solving and technical skills

---

# 42. QC Data Analyst

### Job Summary

The QC Data Analyst analyzes laboratory and production data to identify trends, improve quality processes, and support decision-making.

### Key Responsibilities

- Collect and analyze QC data from multiple sources
- Perform statistical analysis and trend evaluation
- Create dashboards and reports (KPIs, quality metrics)
- Identify deviations, patterns, and process improvements
- Support data-driven decision-making
- Ensure data integrity and accuracy

### Qualifications

- B.Sc / M.Sc / B.Tech in Data Science, Statistics, or related field

### Skills Required

- Data analysis tools (Excel, Python, R, Power BI)
- Statistical knowledge
- Data visualization
- Analytical thinking

---

# 43. Automation QC Specialist

### Job Summary

The Automation QC Specialist designs and manages automated systems to improve efficiency, accuracy, and consistency in quality control processes.

### Key Responsibilities

- Implement lab automation systems and robotic workflows

- Integrate instruments with software systems
- Monitor automated testing processes
- Troubleshoot automation and system issues
- Ensure compliance with validation requirements
- Improve efficiency through process automation

### Qualifications

- B.Tech / M.Tech in Engineering, Automation, or related field

### Skills Required

- Knowledge of automation tools and robotics
- Programming basics (Python, PLC, etc.)
- System integration
- Problem-solving skills

---

# 44. AI/ML Quality Analyst

### Job Summary

The AI/ML Quality Analyst applies artificial intelligence and machine learning techniques to enhance quality control processes, predictive analytics, and anomaly detection.

### Key Responsibilities

- Develop and apply ML models for quality prediction and anomaly detection
- Analyze large datasets from QC labs and manufacturing
- Build predictive models for defect detection and process optimization
- Collaborate with QC, QA, and data science teams
- Validate AI/ML models for regulatory compliance
- Create reports and dashboards for insights

### Qualifications

- B.Tech / M.Tech / M.Sc in AI, Data Science, Computer Science, or related field

### Skills Required

- Machine learning (Python, TensorFlow, Scikit-learn)
- Data analysis and statistics
- Knowledge of QC processes
- Problem-solving and innovation

# 45. QA Analyst (from QC Background)

### Job Summary

The QA Analyst ensures that all quality systems, processes, and documentation comply with regulatory standards. Candidates with QC experience bring strong technical and testing knowledge into QA operations.

### Key Responsibilities

- Review QC data, test reports, and batch records
- Ensure compliance with GMP, GLP, and SOPs
- Perform line clearance and in-process checks
- Handle deviations, OOS, and CAPA processes
- Support audits and regulatory inspections
- Approve/reject materials and finished products

### Qualifications

- B.Sc / M.Sc / B.Pharm / M.Pharm
- Experience in QC is preferred

### Skills Required

- Strong GMP knowledge
- Documentation and review skills
- Attention to detail
- Cross-functional coordination

---

# 46. Validation Specialist

### Job Summary

The Validation Specialist ensures that processes, equipment, and analytical methods consistently produce results meeting predetermined specifications.

### Key Responsibilities

- Perform process, cleaning, and method validation
- Prepare validation protocols (IQ, OQ, PQ)
- Execute validation studies and document results
- Analyze validation data and generate reports
- Ensure compliance with regulatory guidelines

- Support audits and inspections

### Qualifications

- B.Pharm / M.Pharm / B.Sc / Engineering

### Skills Required

- Knowledge of validation lifecycle
- Analytical and documentation skills
- Regulatory compliance understanding

---

# 47. Qualification Engineer

### Job Summary

The Qualification Engineer is responsible for qualifying equipment, utilities, and facilities to ensure they meet operational and regulatory requirements.

### Key Responsibilities

- Perform equipment qualification (IQ, OQ, PQ)
- Prepare qualification protocols and reports
- Ensure proper installation and operation of equipment
- Conduct risk assessments
- Coordinate with engineering and QA teams
- Maintain qualification documentation

### Qualifications

- B.Tech / M.Tech in Engineering (Mechanical, Electrical, Instrumentation)

### Skills Required

- Knowledge of equipment and systems
- Validation and qualification expertise
- Problem-solving and technical skills

---

# 48. Deviation & CAPA Specialist

### □ Job Summary

The Deviation & CAPA Specialist manages quality deviations and implements corrective and preventive actions to improve processes and ensure compliance.

### Key Responsibilities

- Investigate deviations and non-conformances
- Perform root cause analysis (RCA)
- Develop and implement CAPA plans
- Track CAPA effectiveness
- Maintain deviation and CAPA records
- Support audits and inspections

### Qualifications

- B.Sc / M.Sc / B.Pharm / M.Pharm

### Skills Required

- Root cause analysis tools (5 Why, Fishbone)
- Analytical thinking
- Documentation and compliance
- Problem-solving skills

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# 49. Change Control Analyst

### Job Summary

The Change Control Analyst manages and evaluates changes in processes, systems, and documentation to ensure controlled implementation and compliance.

### Key Responsibilities

- Initiate and manage change control processes
- Assess impact and risk of proposed changes
- Coordinate approvals from cross-functional teams
- Ensure proper documentation and implementation
- Track change effectiveness
- Maintain change control records

### Qualifications

- B.Sc / M.Sc / B.Pharm / Engineering

### Skills Required

- Knowledge of change management systems

- Risk assessment skills
- Documentation and coordination
- Regulatory awareness

# 50. Senior QC Scientist

### Job Summary

The Senior QC Scientist leads advanced analytical and scientific activities, ensuring high-quality testing, method reliability, and regulatory compliance.

### Key Responsibilities

- Perform and oversee complex analytical testing
- Lead method development, validation, and transfer
- Review and approve analytical data and reports
- Investigate OOS, deviations, and root causes
- Mentor junior scientists and analysts
- Support audits and regulatory inspections

### Qualifications

- M.Sc / PhD in Chemistry, Biotechnology, or related field
- 5–8 years of experience

### Skills Required

- Advanced analytical expertise
- Leadership and mentoring
- Strong problem-solving ability
- Regulatory knowledge

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# 51. QC Team Lead / Supervisor

### Job Summary

The QC Team Lead manages daily laboratory operations, supervises staff, and ensures timely and compliant testing.

### Key Responsibilities

- Supervise QC analysts and assign tasks
- Monitor lab performance and productivity
- Ensure adherence to GMP, GLP, and SOPs
- Review test results and reports
- Handle deviations, CAPA, and investigations
- Train and develop team members

### Qualifications

- B.Sc / M.Sc / B.Pharm
- 4–7 years of experience

### Skills Required

- Team management and leadership
- Time management
- Decision-making skills
- Strong communication

---

# 52. QC Manager

### Job Summary

The QC Manager oversees all quality control operations, ensuring compliance with regulatory standards and efficient laboratory management.

### Key Responsibilities

- Manage QC laboratory operations and staff
- Develop and implement quality control strategies
- Ensure compliance with GMP, regulatory requirements
- Review and approve analytical results and reports
- Manage budgets, resources, and equipment
- Lead audits and regulatory inspections
- Coordinate with QA, production, and R&D teams

### Qualifications

- M.Sc / B.Pharm / M.Pharm / PhD
- 8–12 years of experience

### Skills Required

- Leadership and strategic planning
- Regulatory expertise
- Resource and project management

- Problem-solving and decision-making

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# 53. Head of Quality Control

### Job Summary

The Head of QC is responsible for overall quality control strategy, laboratory operations, compliance, and continuous improvement across the organization.

### Key Responsibilities

- Define QC policies, goals, and strategies
- Oversee all QC labs and operations
- Ensure global regulatory compliance (USFDA, EMA, WHO)
- Lead audits, inspections, and regulatory interactions
- Drive quality improvements and innovation
- Manage senior teams and cross-functional coordination
- Ensure data integrity and quality systems effectiveness

### Qualifications

- M.Sc / PhD / M.Pharm
- 12–18+ years of experience

### Skills Required

- Strategic leadership
- Deep regulatory knowledge
- Decision-making at organizational level
- Strong communication and stakeholder management

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# 54. Director – Quality

### Job Summary

The Director – Quality is a top-level executive responsible for overall quality strategy, governance, compliance, and global quality systems across the organization.**Key Responsibilities**

- Define and implement global quality strategy
- Oversee QC, QA, validation, and compliance functions
- Ensure alignment with international regulatory standards

- Lead high-level regulatory interactions and audits
- Drive digital transformation and quality innovation
- Manage budgets, risk, and organizational quality culture
- Report to executive leadership and stakeholders

### Qualifications

- PhD / M.Pharm / M.Sc
- 15–20+ years of experience

### Skills Required

- Executive leadership
- Strategic vision and planning
- Global regulatory expertise
- Risk management and governance
- Change management

# 55. Pharmaceutical QC Analyst

### Job Summary

The Pharmaceutical QC Analyst is responsible for testing and analyzing raw materials, in-process samples, and finished pharmaceutical products to ensure they meet quality standards and regulatory requirements.

### Key Responsibilities

- Perform analytical testing (assay, dissolution, content uniformity, impurity profiling)
- Analyze raw materials, intermediates, and finished dosage forms (tablets, capsules, injectables, syrups)
- Operate analytical instruments (HPLC, GC, UV, FTIR)
- Prepare and standardize reagents and solutions
- Document test results and maintain laboratory records
- Investigate out-of-specification (OOS) and out-of-trend (OOT) results
- Follow GMP, GLP, and pharmacopeial standards (USP, IP, BP)
- Support method validation and transfer activities

### Qualifications

- B.Pharm / M.Pharm / B.Sc / M.Sc in Chemistry or related field

### Skills Required

- Strong knowledge of pharmaceutical analysis
- Instrument handling and troubleshooting
- Attention to detail
- Documentation and compliance skills

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# 56. Stability Program Manager

### Job Summary

The Stability Program Manager is responsible for designing, managing, and overseeing stability studies to ensure the shelf-life, safety, and efficacy of pharmaceutical products in compliance with regulatory guidelines.

### Key Responsibilities

- Design and implement stability programs as per ICH guidelines (Q1A)
- Manage stability chambers (temperature, humidity control)
- Plan and schedule stability studies (long-term, accelerated, intermediate)
- Monitor sample testing at defined intervals
- Review and trend stability data
- Prepare stability reports for regulatory submissions (ANDA, NDA)
- Ensure compliance with global regulatory standards (USFDA, EMA, WHO)
- Lead stability-related audits and inspections
- Coordinate with QC, QA, and regulatory affairs teams

### Qualifications

- M.Pharm / B.Pharm / M.Sc (Chemistry / Pharmacy)
- 6–10+ years of experience (for managerial role)

### Skills Required

- Deep knowledge of ICH stability guidelines
- Data analysis and trending
- Project management and planning
- Regulatory submission experience
- Leadership and coordination

# 57. Food Quality Analyst

### Job Summary

The Food Quality Analyst is responsible for testing and evaluating food products to ensure they meet quality, safety, and regulatory standards before reaching consumers.

### Key Responsibilities

- Perform physical, chemical, and microbiological testing of food products
- Analyze raw materials, in-process samples, and finished food items
- Test parameters such as pH, moisture content, fat, protein, sugar, and acidity
- Conduct sensory evaluation (taste, odor, texture, appearance)
- Maintain laboratory records and quality documentation
- Ensure compliance with food safety standards and SOPs
- Investigate quality deviations and non-conformance issues
- Support product development and quality improvement

### Qualifications

- B.Sc / M.Sc in Food Science, Microbiology, Chemistry, or Biotechnology

### Skills Required

- Knowledge of food testing methods
- Basic microbiology and chemistry understanding
- Attention to detail
- Documentation and reporting skills

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# 58. Food Safety QC Specialist

### Job Summary

The Food Safety QC Specialist ensures that all food manufacturing processes comply with food safety regulations and international standards to prevent contamination and ensure consumer safety.

### Key Responsibilities

- Implement and monitor food safety management systems (HACCP, ISO 22000, FSSAI)
- Conduct hazard analysis and risk assessments
- Monitor hygiene practices in production areas
- Perform environmental and microbial testing in food plants
- Ensure compliance with sanitation and GMP standards
- Investigate food safety incidents and contamination cases
- Conduct internal audits and inspections
- Train staff on food safety procedures and hygiene practices

- Maintain documentation for regulatory audits

### Qualifications

- B.Sc / M.Sc in Food Technology, Microbiology, or Life Sciences

### Skills Required

- Strong knowledge of HACCP, GMP, ISO 22000
- Food microbiology and safety systems
- Risk assessment and auditing skills
- Communication and training ability

# 59. Cosmetic QC Analyst

## Job Summary

The Cosmetic QC Analyst is responsible for testing raw materials, in-process samples, and finished cosmetic products to ensure they meet quality, safety, stability, and regulatory requirements before reaching consumers.

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Key Responsibilities

## Laboratory Testing

- Perform physical and chemical testing of cosmetic products (creams, lotions, shampoos, perfumes, etc.)
- Test parameters such as:
    - pH level
    - Viscosity
    - Color and odor consistency
    - Emulsion stability
    - Active ingredient concentration
- Conduct microbial limit testing for contamination control

---

## Raw Material & In-Process Testing

- Analyze incoming raw materials (oils, surfactants, fragrances, preservatives)
- Verify compliance with approved specifications
- Perform in-process checks during production batches
- Ensure batch consistency and uniformity

## Stability & Shelf-Life Testing

- Conduct stability studies under different conditions (temperature, humidity, light exposure)
- Monitor product changes over time (separation, discoloration, odor changes)
- Support shelf-life determination

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## Documentation & Compliance

- Maintain laboratory notebooks and electronic records
- Prepare Certificates of Analysis (CoA)
- Ensure compliance with GMP, ISO, and cosmetic regulations
- Document deviations, non-conformances, and corrective actions

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## Quality Assurance Support

- Support internal audits and regulatory inspections
- Coordinate with QA and production teams
- Assist in root cause analysis for product failures
- Ensure adherence to SOPs and hygiene standards

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### Qualifications

- B.Sc / M.Sc in Chemistry, Microbiology, Cosmetic Science, or Biotechnology
- Diploma in Cosmetic Technology (preferred in some industries)

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### Skills Required

- Knowledge of cosmetic formulation and ingredients
- Understanding of microbial testing and contamination control
- Analytical and problem-solving skills
- Attention to detail and documentation accuracy
- Familiarity with cosmetic regulations (FDA, EU Cosmetics Regulation, BIS standards)

# 60. Chemical QC Analyst

## Job Summary

The Chemical QC Analyst is responsible for testing raw materials, intermediates, and finished chemical products to ensure they meet quality specifications, safety standards, and regulatory requirements.

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### Key Responsibilities

## Laboratory Testing

- Perform chemical analysis of raw materials and finished products
- Conduct assays, purity testing, and impurity profiling
- Perform titration, gravimetric, and volumetric analysis
- Analyze pH, conductivity, viscosity, and density of chemical samples
- Operate instruments such as:
    - HPLC
    - GC (Gas Chromatography)
    - UV-Vis Spectrophotometer
    - FTIR

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## Raw Material & Process Control

- Test incoming raw materials for compliance with specifications
- Monitor in-process chemical reactions and parameters
- Verify batch consistency and product quality
- Approve or reject materials based on test results

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## Documentation & Reporting

- Maintain laboratory notebooks and electronic records
- Prepare Certificates of Analysis (CoA)
- Document deviations and non-conformances
- Ensure compliance with GMP and ISO standards

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## Quality & Compliance Support

- Support internal audits and inspections
- Assist in root cause analysis for failed batches
- Follow SOPs and safety guidelines strictly
- Ensure chemical handling and storage compliance

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## Qualifications

- B.Sc / M.Sc in Chemistry, Industrial Chemistry, or Applied Chemistry

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## Skills Required

- Strong analytical chemistry knowledge
- Instrument handling (HPLC, GC, UV, FTIR)
- Attention to detail and accuracy
- Understanding of chemical safety (MSDS handling)
- Documentation and compliance skills

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# 61. Polymer Quality Analyst

### Job Summary

The Polymer Quality Analyst is responsible for testing and evaluating polymer materials (plastics, resins, rubbers) to ensure mechanical, thermal, and chemical properties meet required standards.

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## Key Responsibilities

### Polymer Testing & Analysis

- Test physical and mechanical properties of polymers:
  - Tensile strength
  - Elongation
  - Hardness
  - Impact resistance
- Conduct thermal analysis (melting point, glass transition temperature)
- Perform rheological testing (flow and viscosity behavior)
- Analyze molecular weight distribution and composition

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### Material Quality Control

- Inspect raw polymer materials (resins, additives, fillers)
- Monitor production batches for consistency
- Evaluate defects such as cracking, warping, or discoloration
- Ensure compliance with product specifications

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## Instrumentation & Tools

- Operate instruments such as:
  - Universal Testing Machine (UTM)
  - DSC (Differential Scanning Calorimetry)
  - TGA (Thermogravimetric Analysis)
  - FTIR
  - Rheometers

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## Documentation & Reporting

- Maintain test records and quality reports
- Prepare batch release documentation
- Record deviations and initiate investigations
- Support ISO and ASTM compliance requirements

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## Quality & Process Support

- Assist in process optimization for polymer production
- Support troubleshooting of production defects
- Participate in supplier material evaluation
- Support audits and customer quality complaints

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### Qualifications

- B.Sc / M.Sc in Polymer Science, Chemistry, Materials Science, or Chemical Engineering

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### Skills Required

- Knowledge of polymer chemistry and material science
- Instrument handling and mechanical testing expertise
- Analytical thinking and problem-solving
- Understanding of ASTM/ISO material standards
- Strong documentation skills

# 62. QC Consultant

## Job Summary

A QC Consultant provides expert guidance to organizations on quality control systems, laboratory operations, compliance, and process improvements on a freelance or contract basis.

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### Key Responsibilities

- Review and improve QC laboratory systems and workflows
- Design or optimize quality control procedures and SOPs
- Provide technical expertise for troubleshooting OOS/OOT issues
- Support companies in regulatory compliance (GMP, GLP, ISO)
- Train QC staff on best practices and analytical techniques
- Assist in audit preparation and regulatory inspections
- Recommend improvements in lab efficiency and data integrity

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### Qualifications

- M.Sc / M.Pharm / PhD in relevant science field
- 5–15+ years of QC industry experience

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### Skills Required

- Strong analytical and technical expertise
- Deep understanding of regulatory frameworks
- Problem-solving and consulting mindset
- Communication and training ability

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# 63. GMP Auditor

## Job Summary

A GMP Auditor evaluates manufacturing and laboratory systems to ensure compliance with Good Manufacturing Practices and regulatory standards.

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### Key Responsibilities

- Conduct internal and external GMP audits
- Evaluate compliance with SOPs, FDA, EMA, WHO guidelines
- Identify deviations, non-compliance, and risks
- Prepare audit reports with observations and recommendations
- Verify CAPA (Corrective and Preventive Actions) implementation
- Assess documentation, records, and data integrity systems

- Conduct supplier/vendor audits

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### Qualifications

- B.Pharm / M.Pharm / M.Sc / Engineering background
- Certification in auditing (preferred: ISO, GMP, Six Sigma)

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### Skills Required

- Strong knowledge of GMP regulations
- Risk assessment and critical thinking
- Audit reporting and documentation
- Attention to detail and integrity

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# 64. Method Validation Consultant

## Job Summary

The Method Validation Consultant specializes in designing, executing, and reviewing analytical method validation studies for regulatory approval.

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### Key Responsibilities

- Develop and validate analytical methods (HPLC, GC, UV, etc.)
- Design validation protocols (accuracy, precision, specificity, linearity, robustness)
- Review and approve validation reports
- Support method transfer between labs or sites
- Ensure compliance with ICH Q2 guidelines
- Troubleshoot analytical method failures
- Support regulatory submissions (ANDA, NDA, DMF)

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### Qualifications

- M.Sc / M.Pharm / PhD in Chemistry or Pharmaceutical Sciences
- Strong analytical laboratory experience (5–12+ years)

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### Skills Required

- Expertise in analytical chemistry
- Regulatory knowledge (ICH, FDA, EMA)

- Statistical analysis and data interpretation
- High-level documentation skills

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# 65. Quality Compliance Freelancer

## Job Summary

A Quality Compliance Freelancer works independently on short-term projects related to audits, documentation, regulatory compliance, and quality system improvements.

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### Key Responsibilities

- Prepare and review SOPs, quality manuals, and validation documents
- Assist companies in GMP/ISO compliance projects
- Conduct gap analysis of quality systems
- Support regulatory submissions and inspections
- Provide remote consultation for quality issues
- Help implement CAPA and deviation management systems
- Assist startups in setting up QC/QA systems

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### Qualifications

- B.Sc / M.Sc / B.Pharm / M.Pharm / Engineering background
- Strong industry experience preferred (3–10+ years)

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### Skills Required

- Strong documentation and regulatory knowledge
- Independent working ability
- Analytical and problem-solving mindset
- Communication and consulting skills